The next generation of APEX models for the discovery and design of safe antimicrobial peptides

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Antimicrobial peptides (AMPs) offer a promising avenue for the next generation of antibiotics, but identifying potent, selective and safe candidates is challenging. Predictive deep learning models have revolutionized the field and greatly accelerated the identification and design of potent AMPs, however, their performance remains modest. Here we present three new frontier models for antimicrobial activity, cytotoxicity and hemolysis prediction: APEX2, ApexTox and ApexHemo. These models combine high-quality curated in-house and public datasets, modern architectures and revised training formulations. APEX2 outperforms the evaluated models in activity discrimination and potency ranking, while ApexTox and ApexHemo outperform the evaluated public models in toxicity discrimination. Together, these models help prioritize peptides within a predicted therapeutic window, supporting efforts to discover the next generation of antibiotics.

1 Introduction

Peptides are ubiquitous throughout the tree of life, playing important roles in various biological processes. Despite being comparatively small relative to proteins, the combinatorial molecular space they inhabit is large enough to support diverse functions, from signaling and molecular recognition to host defense. They also occupy a favorable middle ground between larger biologics

and small molecules, combining extensive chemical diversity with a small molecular size and ease of synthesis, making them an attractive drug modality [1]. One of the most promising applications of peptides is in the development of next-generation antimicrobial agents to address antimicrobial resistance. Antimicrobial peptides (AMPs) play an important role in the innate immune system of many organisms and are believed to constitute one of its most ancient parts. These peptides have evolved over millions of years to effectively fend off bacteria, making them excellent starting points for novel antibiotics [2].

Computational methods have emerged as a cornerstone in efforts to identify and design peptide-based therapeutics, particularly AMPs [3]. These efforts can be divided into three categories: mining, *de novo* design and template optimization. Mining identifies promising sequences within existing proteomes, unveiling a molecular layer of encrypted peptides embedded in the proteomes of organisms [4]. In contrast, *de novo* design seeks to generate new sequences unseen in nature [5], while template optimization seeks to improve the therapeutic quality of a starting template peptide [6]. Predictive models are the foundational engines of these methods, acting as proxies for experimental outcomes and thereby enabling both large-scale mining efforts and rapid development. Several computational approaches have been developed to identify AMPs and predict their potency, ranging from simple physicochemical and sequence-based heuristics to modern deep learning models. Examples of well-used machine learning approaches include AMPlify [7], amPEPpy [8], HydrAMP [9], OmegAMP [5], AMPGen [10] and others, such as our previously published APEX [4]. These models learn from experimentally tested sequence–activity relationships, often using public datasets such as DBAASP [11], DRAMP [12] and APD [13], which are assembled from years of experimental efforts distributed across research groups.

These efforts have led to the identification and design of several highly potent AMPs targeting notoriously hard-to-treat bacterial strains, including those exhibiting resistance towards conventional antibiotics, yet translating them into clinically useful drugs remains challenging. This is partly due to the fact that antimicrobial activity and host-cell toxicity can arise from related membrane interactions, and identifying or designing potential drug candidates therefore requires considering these hemolytic and cytotoxic effects alongside antimicrobial potency [14]. This trade-off is captured in the therapeutic window in which antimicrobial activity is achieved below concentrations that damage host cells [15].

Despite progress in the field, the prediction of both AMP activity and toxicity remains modest. Models tend to capture broad physicochemical trends, suffer from systematic biases, and only work well within narrow sequence domains but exhibit limited performance on out-of-distribution benchmarks. Moreover, most AMP models do not predict strain-specific potencies, and therefore model general AMP-like features [16]. We reason that many of these flaws stem largely from poor data annotation, and modelling assumptions and practices, leaving room for improvement in both domains. Importantly, improving these models could enable more accurate screening during mining campaigns, allowing us to explore the encrypted innate immune repertoire more deeply, and support the design of more effective therapeutics.

Here, we advance computational AMP discovery by addressing both antimicrobial potency and safety with three new state-of-the art models: APEX2, ApexTox and ApexHemo. To train these models, we create a dataset combining extensively curated, high-quality in-house experimental data with curated public datasets. We then revisit the underlying formulations and modelling assumptions behind the training of these models, deviating from the common paradigms in the field on theoretical grounds. Lastly, we combine these efforts to develop and train models using modern deep learning architectures. Overall, these models outperform current alternatives. We combine these models with other APEX peptide-property predictors in ApexSuite to facilitate their use in mining and design workflows.

2 Results

Across more than 50 projects, our laboratory has tested both naturally occurring and designed antimicrobial peptides (AMPs) under standardized experimental conditions against 11 ESKAPEE strains (**Fig. 1a**). These measurements supported the development of the original APEX-models [4], which remain competitive among AMP prediction models [16]. Here, we develop sought to APEX2, the next generation in this series, offering improved capabilities and features. In addition to expanding the dataset, we revise the core formulation underlying AMP prediction and incorporate modern representation learning methods into the model development pipeline.

2.1 The data and model assumptions underlying APEX2

Since the release of APEX, our in-house dataset has continued to expand alongside the growth of public datasets and now contains 37,085 observed peptide–strain targets across 3,691 peptides (**Fig. 1a**). To support strain-specific prediction, we combined these in-house assays with high-quality public MIC records matched to the strains in our panel, yielding 38,492 peptide–strain targets across 4,930 peptides. We also assembled a broader activity dataset from public AMP records and sampled UniProt fragments [17], providing binary activity labels for 49,188 peptides and extending sequence coverage beyond the strain-specific assays (**Fig. 1b**). Accounting for overlap between these tasks, the dataset contains 52,918 distinct sequences.

When revisiting APEX, we identified two limitations in how common prediction formulations represent experimental measurements. Firstly, models commonly frame prediction as either a classification or a regression problem. In the classification formulation, the goal is to distinguish AMPs from non-AMPs or identify peptides that meet a defined potency threshold [9]. In the regression formulation, the task is to predict the experimentally measured minimum inhibitory concentration (MIC). In conventional regression, inactive measurements are often excluded or assigned a high MIC through imputation.

However, modelling experimentally measured antimicrobial potency requires jointly addressing two related questions: whether a peptide inhibits growth within the tested concentration range, and, if it does, at what concentration. Classification retains the first question but discards differences in potency, while regression with imputed inactive values combines the two through an arbitrarily assigned MIC. We instead formulate APEX2 as a two-part hurdle model that jointly predicts strain-specific activity and MIC conditional on activity (**Fig. 1c**).

Secondly, standard regression is typically performed directly on the reported MIC. However, MICs are measured on a discrete concentration ladder, usually with twofold dilutions. When adjacent tested concentrations permit and inhibit growth, respectively, the MIC identifies the interval between them. Treating the reported value as exact penalizes predictions that fall within this experimentally supported interval. To account for this, we use an interval-based loss, which penalizes the squared distance outside the interval in logarithmic concentration space (**Fig. 1c**). We combine this term with binary cross-entropy losses for broad and strain-specific activity and penalties that encourage similar activity and potency output weights for phylogenetically related strains. Together, this formulation provide a more faithful model of the measured antimicrobial activity.

APEX2 is an ensemble of neural networks that share a three-head design (**Fig. 1d**). Each member has two main output heads: one predicts the probability of activity against each strain, and the other predicts MIC conditional on activity. These heads are supervised by high-quality in-house assays and carefully curated, strain-matched public MIC data. Active and inactive assays train the activity head, while measured MICs from active assays also train the conditional

potency head. A third, broad activity head is trained on public AMP annotations and activity records, supplemented with sampled UniProt fragments as presumed inactive background. It provides a broad AMP probability comparable to those of conventional AMP classifiers. All three heads use a shared sequence representation built with modern embedding methods and architectures, allowing the broader activity evidence to contribute to representation learning.

The final ensemble was selected from a search of 300 hyperparameter configurations spanning encoder architecture, feature combinations, network dimensions and training settings (**Fig. 1e**). The variance among ensemble-member predictions summarizes model disagreement and is available to the user alongside the ensemble mean.

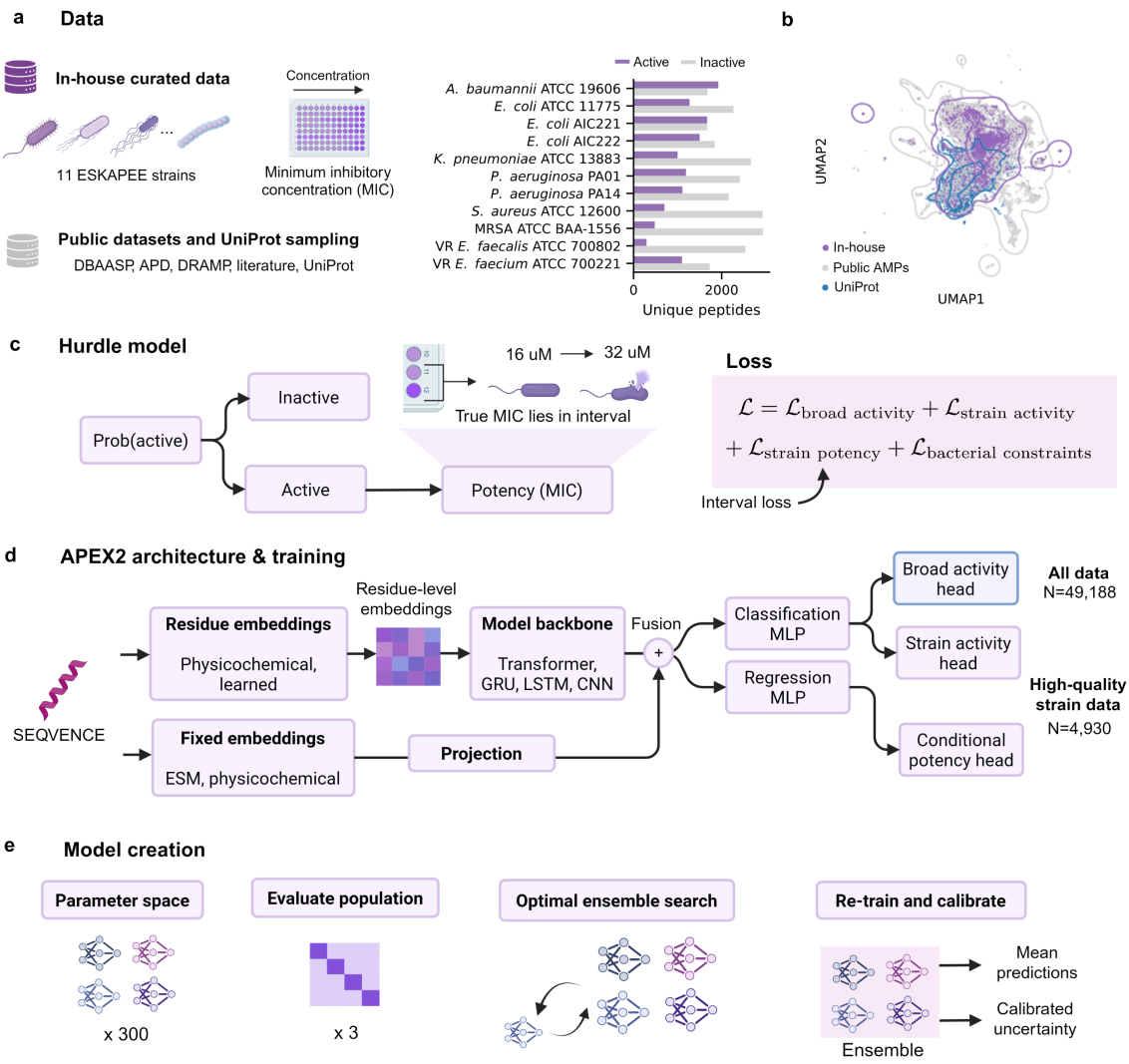


Figure 1: APEX2 data, formulation and architecture. **a,** In-house MIC measurements across 11 ESKAPEE strains were combined with curated public activity records and sampled UniProt fragments. Bars show the number of unique in-house peptides with observed activity (purple) or no inhibition within the tested concentration range (gray) against each strain. **b,** UMAP projection of standardized ESM2-35M sequence embeddings for in-house peptides, public records and sampled UniProt fragments. Each point represents a unique peptide, and contours show the density of each source group. **c,** The hurdle formulation separates the probability of activity from MIC conditional on activity. In the illustrated twofold dilution, growth at 16 μM and inhibition at 32 μM constrain the MIC to $(16, 32]$ μM. The conditional potency loss is zero within this interval and penalizes squared distance outside it in logarithmic concentration space. Binary cross-entropy terms train broad and strain-specific activity, while bacterial constraints encourage similar strain-specific output weights for phylogenetically related targets. Only active assays supervise conditional potency. **d,** Residue-level physicochemical and learned embeddings are processed by a CNN, recurrent network or Transformer and combined with optional frozen ESM2 and physicochemical sequence features. The shared representation feeds broad activity, strain-specific activity and conditional MIC heads. The main-head dataset comprises 4,930 peptides from in-house assays and high-quality public MIC records across all data partitions; the broad activity dataset comprises 49,188 peptides. **e,** Three-fold cross-validation evaluates 300 hyperparameter configurations, followed by ensemble selection using out-of-fold activity discrimination and potency ranking. The 14 selected members are refitted on the training data, with validation-based checkpoint selection and calibration of strain activity probabilities and potency uncertainty.

2.2 APEX2 improves activity prediction and potency ranking

To train APEX2, the in-house data were divided into sequence similarity-aware training, validation and test sets containing 2,584, 738 and 369 peptides, respectively (70/20/10%). The validation set guided checkpoint selection and calibration, while the test set was reserved for final evaluation. The model development pipeline produced an ensemble of 14 members comprising six CNNs, five bidirectional LSTMs, two bidirectional GRUs and one Transformer, with different combinations of learned, physicochemical and protein language model embeddings. Below, we first compare the performance of APEX2 with other models and then characterize its outputs.

APEX2 improved both activity prediction, potency ranking and absolute prediction relative to the compared models (**Fig. 2a–c**; **Extended Data Fig. 1a**; **Table 1**). On the shared test cohort of 195 peptides, its strain activity head achieved the highest mean AUROC and AUPRC, while its conditional MIC head achieved the highest interval concordance. This remained true without strain-level resolution (**Extended Data Table 1**). Several classifiers identified a large fraction of active peptides but also assigned activity to many inactive ones. OmegAMP had the lowest false-positive rate, at the cost of the lowest true-positive rate, consistent with its design and purpose [5]. It should be noted that our test set is relatively difficult, as its peptides were originally selected for their potential antimicrobial activity, making the task one of distinguishing activity among plausible candidates. APEX2 also achieved the highest interval concordance on the sequence-filtered external benchmark [18], extending the improvement in potency ranking beyond the in-house data (**Fig. 2d**).

Table 1: Antimicrobial activity discrimination and potency prediction on the in-house test cohorts.

Model / head	AUPRC	AUROC	FPR (%)	TPR (%)	LR+	Prec@100 (%)	Interval C	MIC R^2
Shared test cohort: 195 peptides								
ApexPathogen	0.49	0.62	57.76	71.54	1.24	42.45	0.55	−2.22
AMPGen	0.52	0.69	—	—	—	46.27	0.60	−0.59
AMP-READ [19]	0.48	0.65	—	—	—	45.09	0.59	—
OmegAMP	0.57	0.67	**2.82**	15.45	**5.47**	44.64	0.60	—
HydrAMP-AMP	0.47	0.61	87.12	92.02	1.06	41.00	0.59	—
HydrAMP-MIC	0.52	0.66	63.16	81.86	1.30	46.09	0.60	—
AMPlify	0.51	0.62	84.46	92.15	1.09	42.09	0.56	—
amPEPpy	0.48	0.63	84.24	94.55	1.12	43.73	0.55	—
APEX2 strain activity	**0.67**	**0.81**	24.56	62.32	2.54	**52.64**	0.67	—
APEX2 broad activity	0.58	0.73	92.26	**99.56**	1.08	48.73	0.63	—
APEX2 conditional MIC	0.60	0.73	—	—	—	49.27	**0.69**	**0.19**
Complete test cohort: 369 peptides								
APEX2 strain activity	0.69	0.83	14.42	56.66	3.93	63.73	0.65	—
APEX2 broad activity	0.61	0.78	67.11	94.71	1.41	58.73	0.63	—
APEX2 conditional MIC	0.59	0.76	—	—	—	56.27	0.73	0.27

AUPRC is calculated as average precision. All metrics give equal weight to each strain. FPR and TPR are false-positive and true-positive rates; Prec@100 is precision among the 100 highest-ranked peptides per strain. LR+ is the ratio of the unrounded mean TPR to the unrounded mean FPR. Activity metrics use each strain's observed active and inactive assays. AMPlify, amPEPpy, both HydrAMP classifiers, AMP-READ and the APEX2 broad head supply one score per peptide, reused across strains. OmegAMP uses a matching strain classifier, then a species classifier, then its broad classifier. Higher classifier probabilities, lower predicted MICs and lower native AMP-READ scores indicate greater activity. HydrAMP-MIC predicts the probability of a low MIC against *E. coli*. Interval concordance (C) evaluates all models as rankers of measured active assays, comparing definitely ordered experimental MIC intervals and giving half credit to prediction ties. R^2 compares predicted and reported numerical MICs in $\log_2$ space for models with quantitative MIC outputs. Binary cutoffs are probability ≥ 0.5 for APEX2 and OmegAMP, > 0.5 for AMPlify, amPEPpy and HydrAMP, and MIC ≤ 128 μM for ApexPathogen. Dashes indicate outputs without the applicable cutoff or quantitative MIC prediction. AMPGen uses its *E. coli* or *S. aureus* output for those species and the mean of its two log-MIC outputs for other species. Bold marks the highest value for each metric, or the lowest FPR, within the shared cohort.

To separate the contributions of the model architecture and training formulation, we trained three additional variants: one retaining the earlier architecture family, one replacing the hurdle and interval formulation with imputed MIC regression, and a baseline retaining both earlier choices. Expanding the range of architectures and changing the formulation each improved on the baseline, while combining them produced the strongest performance (**Fig. 2e**).

The formulation also affected the accuracy of the predicted MICs. Although the non-hurdle models retained some ability to rank active peptides, their predictions showed a systematic offset and negative mean strain-level R^2, compared with positive values for the hurdle models (**Fig. 2f**). Subtracting the mean prediction offset for each strain reduced the error but did not close the gap entirely, and the reformulation therefore reduced both systematic offset and residual prediction error.

The three APEX2 outputs captured related but distinct aspects of antimicrobial activity. Inactive peptides generally received lower strain activity probabilities and higher conditional MICs, whereas the broad head often assigned high probabilities even to peptides without observed activity in the assayed panel (**Fig. 2g**). This was reflected in its higher true-positive and false-positive rates (**Table 1**) and persisted in held-out peptides from public datasets (**Extended Data Fig. 1b**).

The variance among ensemble-member predictions was greatest around an activity probability of 0.5, while the spread in predicted MICs generally increased with their magnitude (**Fig. 2h**).

To examine how these differences appear across a large biological sequence space, we applied APEX2 to the human proteome. Scanning 20,416 reviewed canonical proteins at peptide lengths of 8–35 residues yielded 307,883,202 peptide occurrences. We then compared the upper and lower 20% of sampled predictions within each peptide length (**Fig. 2i**). Low conditional MICs were associated with enrichment of aromatic and basic residues and depletion of acidic residues. High strain activity probabilities showed a particularly strong enrichment of lysine and arginine for most Gram-negative targets, while the broad head showed a more pronounced enrichment of cysteine and tryptophan.

The relationships between strain outputs also depended on the prediction task (**Fig. 2k**). Activity probabilities were particularly strongly correlated among the Gram-negative targets, whereas their relationships with the Gram-positive targets were more variable. The two *S. aureus* strains had similar potency profiles, while vancomycin-resistant *E. faecalis* showed the most distinct profile across the panel. These differences support retaining individual strain predictions when selecting peptides for a particular pathogen.

Finally, we compared recovery of 15 known human encrypted antimicrobial peptides absent as exact sequences from the APEX2 training data. To combine strain-specific activity and potency, we ranked peptides by MIC divided by activity probability, so that a low activity probability reduced the priority of an otherwise potent prediction. For example, a conditional MIC of 100 μM and an activity probability of 0.2 give a score of 500. A score below 128 in at least one strain recovered 11 of the 15 peptides with APEX2, compared with four using an ApexPathogen MIC cutoff below 128 μM; four were missed by both models (**Fig. 2j**). APEX2 thus improved recovery in this comparison while providing separate outputs for broad antimicrobial potential, strain-specific activity and conditional potency.

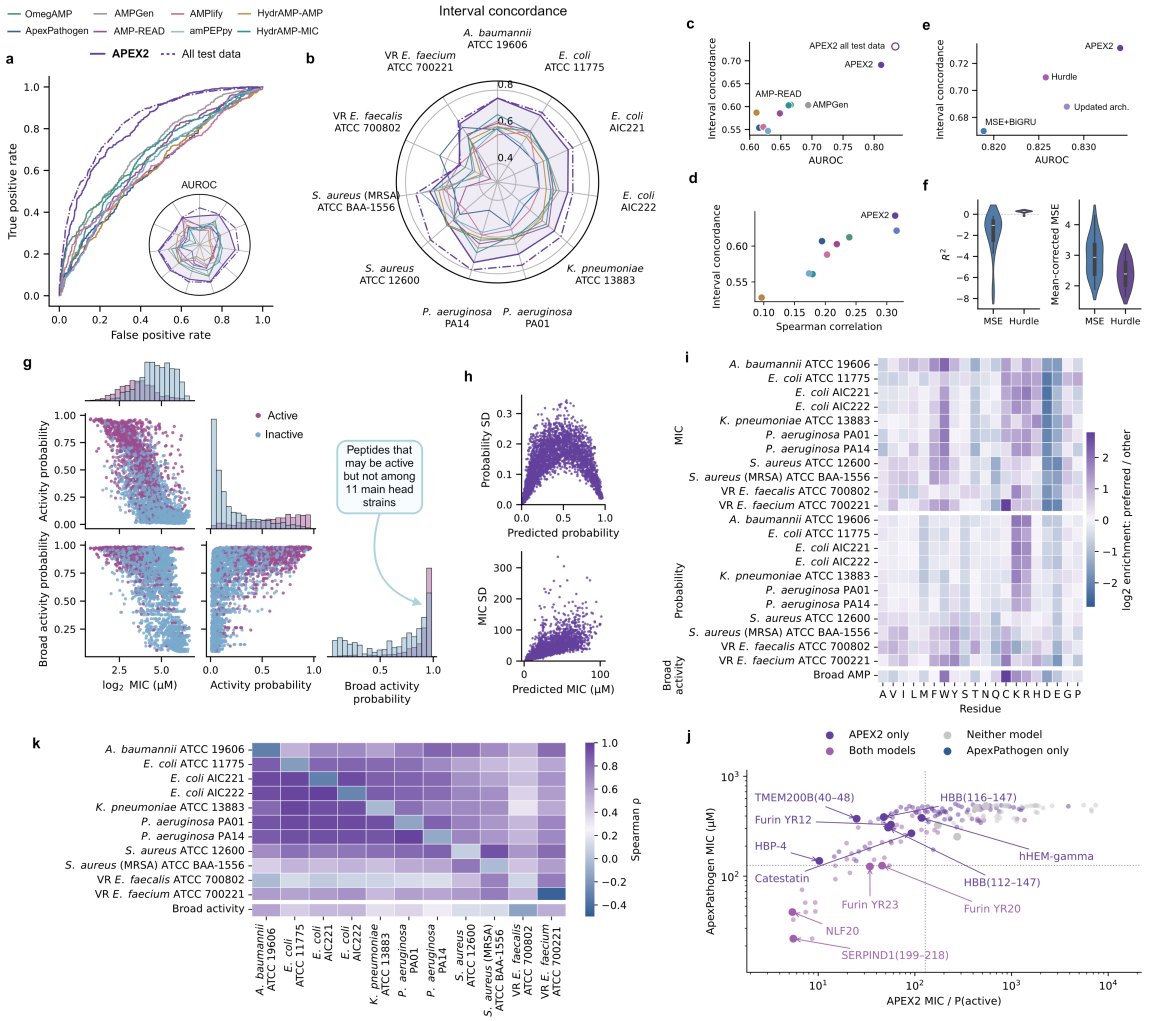


Figure 2: APEX2 performance and human-proteome analysis. **a,** Mean ROC curves and strain-level AUROCs, using the APEX2 strain activity head. Curves show true-positive rates averaged across strains on a common false-positive-rate grid. **b,** Strain-level interval concordance among measured active assays, using APEX2 conditional MICs and each comparator's activity ranking. **c,** Mean AUROC and interval concordance, with equal weight per strain. Higher classifier probabilities and lower MICs or native AMP-READ scores indicate greater activity. **d,** Species-balanced Spearman correlation and interval concordance on the shared external bacterial cohort; output matching is described in Methods. **e,** Architecture and loss ablations on the full test cohort, comparing earlier and expanded encoders under imputed MSE and hurdle/interval training. **f,** R^2 and mean-corrected MSE for the imputed MSE and hurdle formulations. Violins pool strain-level results across architectures; mean correction removes the model–strain mean $\log_2$ prediction error. **g,** Conditional MIC, strain activity probability and broad AMP probability, colored by observed activity. Each point is an assayed peptide–strain pair; marginal histograms show output distributions. **h,** Ensemble standard deviation versus predicted activity probability or MIC. **i,** Log_2 residue-frequency ratios between the preferred and opposite 20% tails, matched by peptide length; lower MICs or higher activity probabilities define the preferred tail. **j,** Recovery of known human encrypted peptides; large points use each model's minimum score across strains. Faint points show individual strains; dashed lines mark the recovery cutoffs of 128 for APEX2 MIC/probability and 128 μM for ApexPathogen MIC. **k,** Spearman correlations among conditional MICs (upper triangle) and activity probabilities (lower triangle), with within-strain MIC–probability correlations on the diagonal. The shared in-house comparison uses 195 test peptides; results for the full APEX2 test set use 369. AMPlify, amPEPpy, both HydrAMP classifiers and AMP-READ reuse one peptide score across strains, evaluated against each strain's labels. OmegAMP uses matching strain, species or broad classifiers, in that order. AMPGen uses matching species predictions or the mean of its two log-MIC outputs.

2.3 Agent-assisted curation consolidates peptide toxicity data

Beyond antimicrobial activity, our laboratory has tested peptides for cytotoxicity and hemolytic activity across 30 projects spanning 2022–2026. Motivated by the success of APEX and the utility of high-quality in-house data, we wanted to curate these additional datasets to train models for safety. However, these measurements remained scattered across project-specific files in different formats. To bring them together for model training, we devised an agent-assisted curation workflow with a human in the loop (**Fig. 3a**). The agents examined raw instrument outputs, native analysis files, processed workbooks, figures and publications, and consolidated the measurements into a common format. Questions about peptide identities, concentrations, normalization and unusual response curves were resolved through human review and consultation. The agents implemented the accepted decisions, and this process continued until the identified issues had been resolved. The resulting dataset then underwent human quality control.

The curation recovered 21,408 cytotoxicity observations and 32,025 hemolysis observations across 1,774 distinct peptide sequences, summarized into 1,700 cytotoxicity curves and 1,474 hemolysis curves. After filtering for model eligibility, the in-house datasets contained 1,612 cytotoxicity assays across 1,464 peptides and 1,407 hemolysis assays across 1,398 peptides (**Fig. 3b**). Cytotoxicity measurements covered five cell contexts, with human embryonic kidney cells (HEK293T) accounting for the largest fraction. Many curves, particularly for hemolysis, did not cross the 50% response level within the tested concentration range. These assays still constrain the endpoint, making retention of censored measurements an important part of the dataset and subsequent modelling.

We complemented the in-house measurements with public quantitative assay records from DBAASP [11] and broader binary toxicity and hemolysis labels from public databases and published datasets (Methods). Their sequence-space distributions overlapped while also covering different regions, broadening the evidence available for training (**Fig. 3b–c**). The agents checked unusual public values against the original experimental reports and identified extrapolated endpoints and assay ceilings recorded as exact measurements. Where supported by the source experiments, these values were corrected to bounds. The resulting collection combines concentration–response measurements, quantitative endpoints and broader binary evidence, preserving the different information each provides about peptide cytotoxicity and hemolysis.

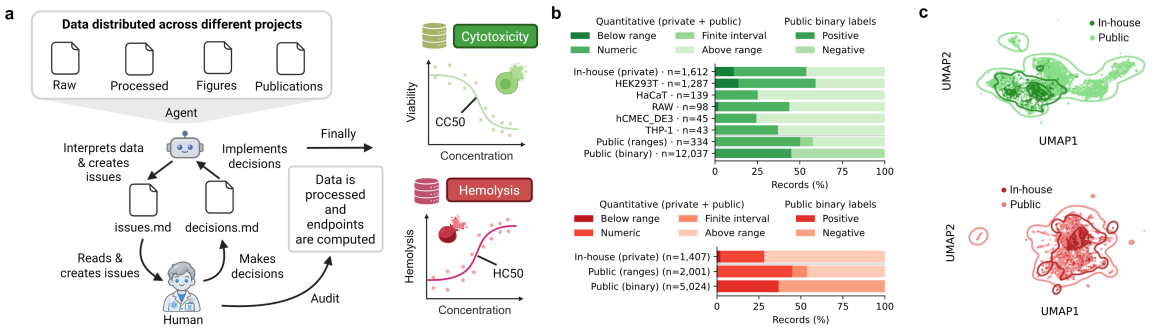


Figure 3: Agentic curation of cytotoxicity and hemolysis data. a, Agent-assisted curation of raw measurements, processed files, figures and publications distributed across projects. Agents extract measurements and flag questions for human review, then implement the accepted decisions in successive rounds of curation. After human review and audit, the data are consolidated into concentration-level responses and CC50/HC50 summaries. CC50 and HC50 denote the concentrations associated with 50% cell viability and hemolysis, respectively. **b,** Composition of the model-ready quantitative and binary datasets. Quantitative records have numerical endpoints, finite intervals, and endpoints below or above the tested range (left- and right-censored, respectively). Public binary records are separated into positive and negative labels. Counts include all data partitions and refer to assays for quantitative data and peptide records for binary data. **c,** UMAP projections of standardized ESM2-35M sequence embeddings for cytotoxicity (top) and hemolysis (bottom). Each point represents a unique peptide, colored by in-house or public origin; contours show the density of each source group.

2.4 ApexTox and ApexHemo predict concentration-dependent toxicity

Building on the design of APEX2 and the newly curated datasets for hemolytic activity and cytotoxicity, we next sought to create models that predict these properties. ApexTox and ApexHemo share the general architecture and model development workflow of APEX2, but their main outputs describe how toxicity changes with concentration (**Fig. 4a**). Each ensemble member predicts a 50% endpoint, CC50 or HC50, together with a fixed or learned Hill slope that defines the concentration–response curve. ApexTox additionally accounts for cell context. A broad activity head learns from eligible binary evidence from in-house and public sources, while the main outputs use curated quantitative measurements.

Even when a response does not cross 50% within the tested range, the measured responses and endpoint bounds provide information about the same concentration–response curve. We therefore retain these assays through censored regression, applying an interval-based loss to exact and bounded endpoints and using measured responses at the assayed concentrations as an additional training signal. The in-house peptides were divided into training, validation and test sets in a 70/20/10% ratio, keeping assays of the same sequence together. Configuration search and ensemble selection produced 11 ApexTox and nine ApexHemo members spanning several architectures and feature combinations.

Below, we first examine agreement with measured toxicity endpoints and compare activity discrimination with contemporary public models, reporting the in-house test and public validation cohorts separately (**Table 2**). For cytotoxicity, we compare ApexTox with ToxinPred3, a classification model trained on toxic and non-toxic peptide labels [20]. In the public data assembled here, binary toxicity annotations greatly outnumbered quantitative cytotoxicity measurements (**Fig. 3b**), motivating the addition of in-house concentration–response data for CC50 modelling. For hemolysis, we compare ApexHemo with HemoPI2's hemolysis classifier and HC50 regressor [21], retrained as described in Methods. We then explore how the predicted toxicity profiles relate to antimicrobial activity and peptide composition.

ApexTox captured a gradient in cytotoxic potency, as peptides with lower measured CC50s generally received response curves shifted toward lower concentrations (**Fig. 4b**). Agreement with numerical endpoints was modest in the in-house test set and stronger in the public validation cohort, while predictions also reflected the ordering of assays with endpoints below, within and above the tested range (**Fig. 4c–d**). This ordering was less apparent in the toxicity scores returned by ToxinPred3. Both the ApexTox endpoint and broad activity outputs achieved higher AUROCs than ToxinPred3 in both cohorts (**Fig. 4e**; **Table 2**). The concentration-dependent output also locates this distinction on a dose scale.

ApexHemo showed a similar relationship between predicted response curves and measured hemolytic potency (**Fig. 4f**). Its HC50 predictions correlated more strongly with numerical measurements than those of HemoPI2 in both cohorts, and more clearly separated assays with endpoints below, within and above the tested range (**Fig. 4g–h**). At an experimental HC50 threshold of 128 μM, both its endpoint and broad activity outputs achieved higher AUROCs than the corresponding HemoPI2 outputs (**Fig. 4i**; **Table 2**). The improvement therefore extended from predicting HC50 to identifying peptides that produce hemolysis within a defined concentration range.

Table 2: Cytotoxicity and hemolysis prediction on in-house test and public validation cohorts.

	In-house test			Public validation		
Model / output	AUROC	Pearson r	C	AUROC	Pearson r	C
Cytotoxicity						
ToxinPred3	0.60	—	0.59	0.61	—	0.59
ApexTox broad	0.75	—	0.72	**0.88**	—	0.77
ApexTox CC50	**0.78**	0.27	**0.76**	0.86	0.74	**0.79**
Hemolysis						
HemoPI2 classifier	0.77	—	0.71	0.87	—	0.78
HemoPI2 HC50	0.70	0.28	0.67	0.81	0.49	0.75
ApexHemo broad	0.91	—	0.86	0.90	—	0.83
ApexHemo HC50	**0.93**	**0.58**	**0.89**	**0.91**	**0.61**	**0.84**

For AUROC, experimental CC50 $\leq$ 50 μM or HC50 $\leq$ 128 μM define positive labels; ambiguous or conflicting classes are excluded. Rankings use continuous classifier or hybrid scores, or the negative $\log_2$ of predicted CC50/HC50, with higher scores indicating greater toxicity. ApexTox CC50 predictions use the assayed cell context; sequence-only ToxinPred3 hybrid scores and ApexTox broad scores are reused across cell contexts. Pearson r compares predicted and measured exact endpoints in $\log_2$ space for quantitative outputs; interval concordance (C) evaluates all models as rankers of definitely ordered exact and censored endpoints. Cytotoxicity labels are resolved within cell context, while concordance pools available contexts. Metrics account for repeated assays by giving equal total weight to each peptide; concordance uses products of these weights, excludes same-sequence pairs and gives half credit to prediction ties. In-house/public peptide counts are 145/64 for cytotoxicity AUROC, 62/36 for r and 146/65 for C; corresponding hemolysis counts are 118/253, 38/138 and 140/284. Public validation contributed to checkpoint selection. HemoPI2 models were retrained as described in Methods. Dashes indicate outputs without a quantitative endpoint. Bold marks the highest value within each task and cohort where multiple outputs are compared.

To explore how these predictions relate to antimicrobial activity, we examined 10,000 human-proteome peptides sampled across lengths and predicted AMP activity groups. ApexTox and ApexHemo endpoints were strongly associated, while their relationships with the APEX2 MIC/probability score were weaker (**Fig. 4j**). Lower antimicrobial scores tended to accompany lower CC50s and HC50s, indicating that peptides predicted to be more potent against bacteria also tended to have greater predicted toxicity in this sample. This association motivates considering both properties when selecting candidates.

The sequence preferences of the toxicity models help explain where these objectives can differ. Peptides with greater predicted toxicity were enriched in leucine and aromatic residues, whereas lysine and arginine were depleted among peptides with high predicted hemolysis (**Fig. 4k**). As in APEX2, the broad and quantitative outputs did not have identical sequence preferences (**Fig. 4l**). CC50-associated enrichments were similar across the five ApexTox cell contexts, with isoleucine, leucine, methionine and aromatic residues enriched among peptides with lower predicted CC50s. Its broad head showed a different profile dominated by cysteine enrichment, consistent with the cysteine enrichment reported among toxic peptides in the ToxinPred3 dataset [20]. ApexHemo's endpoint and broad heads were more similar, sharing enrichment in aromatic residues and cysteine and depletion of lysine and arginine. These patterns further support considering the concentration-dependent outputs alongside broad activity probabilities.

Overall, ApexTox and ApexHemo improve toxicity discrimination relative to the evaluated public models and provide quantitative predictions that reflect the ordering of exact and censored endpoints.

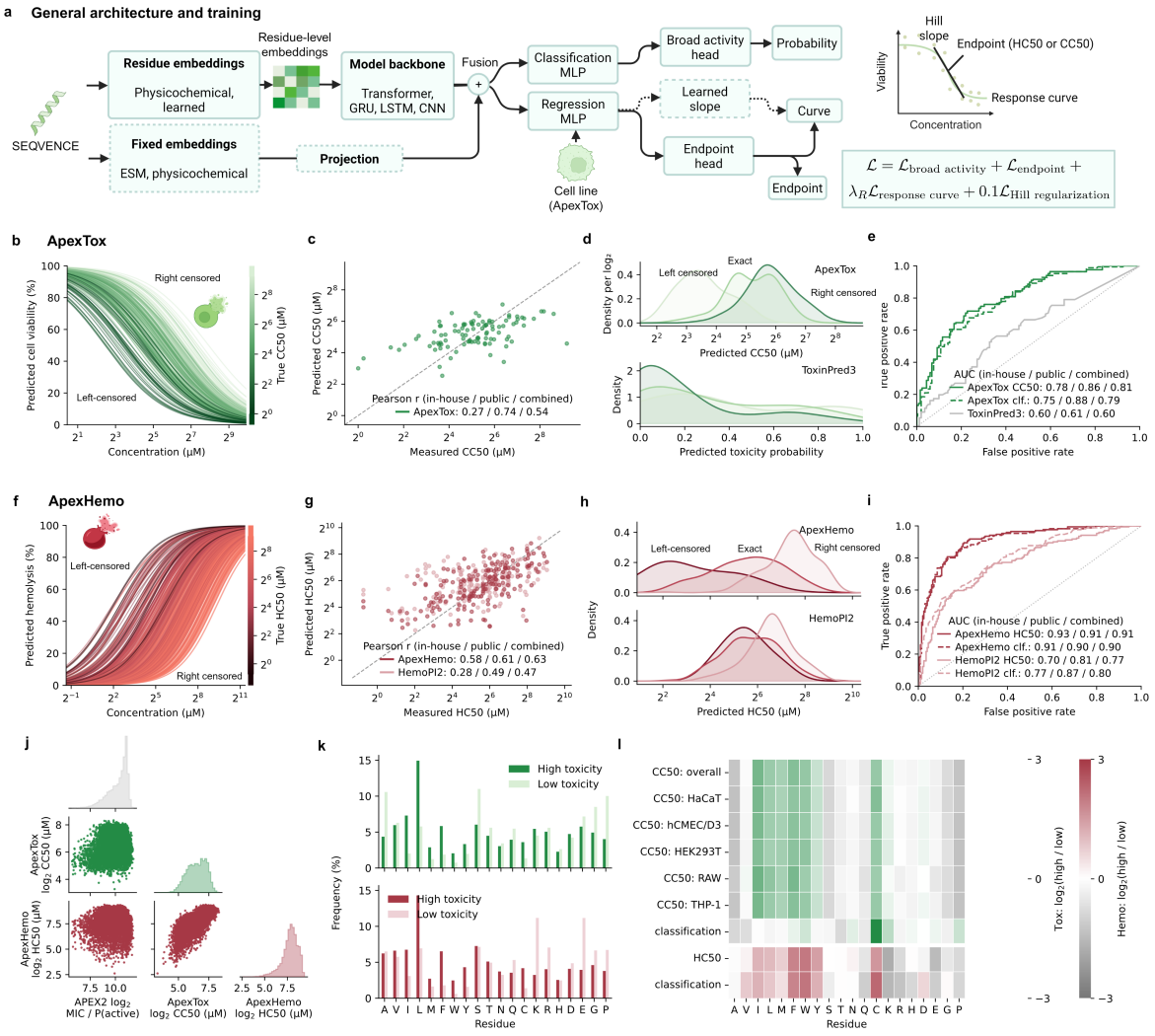


Figure 4: ApexTox and ApexHemo architecture, performance and sequence preferences. **a,** Broad classification and concentration–response prediction, including cell context for ApexTox. **b,** Predicted ApexTox viability curves, colored by experimental CC50 or censoring category. **c,** Predicted CC50 values and exact measurements. Dashed diagonals in c and g indicate agreement between predicted and measured endpoints. **d,** ApexTox CC50 predictions and ToxinPred3 hybrid scores grouped by experimental endpoint censoring. **e,** Toxicity ROC comparison at a CC50 threshold of 50 μM. **f,** Predicted ApexHemo hemolysis curves, colored by experimental HC50 or censoring category. **g,** Comparison with HemoPI2 using exact HC50 measurements. **h,** Predicted HC50 distributions grouped by endpoint censoring. **i,** Hemolysis ROC comparison at an HC50 threshold of 128 μM. **j,** Relationships among antimicrobial and toxicity outputs for human-proteome peptides. Each point represents one of 10,000 sampled peptides. **k,** Residue frequencies in high- and low-toxicity cohorts. **l,** Residue enrichment across toxicity endpoints, broad heads and ApexTox cell contexts. In k–l, high- and low-toxicity cohorts are opposite 20% tails within peptide-length and AMP-score groups; lower endpoints or higher broad probabilities define greater toxicity. Enrichment is the $\log_2$ ratio of residue frequencies in these matched cohorts. Panels b–i combine in-house test and public quantitative-validation cohorts; scatter and ROC annotations give in-house/public/combined metrics, and ROC curves show the combined cohorts. ROC rankings use classifier or hybrid scores or decreasing predicted CC50/HC50. ApexTox CC50 predictions use the assayed cell context, while sequence-only ToxinPred3 and ApexTox broad scores are reused across cell contexts; each peptide has equal total weight. HemoPI2 models were retrained as described in Methods. Groups labelled "Exact" in d and h include exact and finite-interval endpoints. ApexTox uses an unknown cell context in j–k and for the "overall" profile in l.

2.5 ApexSuite enables joint optimization of potency and toxicity

Predictive models play an important role in sequence design, where they can act as surrogates for experiments guiding optimization methods. To illustrate how APEX2, ApexTox and ApexHemo can be used together, we compared design using APEX2 alone with joint design using all three models in an evolutionary optimization process. Optimization with APEX2 alone rapidly improved the antimicrobial score, but also drove the population toward lower cytotoxicity and hemolysis fitness, corresponding to greater predicted toxicity (**Fig. 5a**). Including the two toxicity models maintained improvement in antimicrobial fitness while favoring higher CC50s and HC50s and lower broad toxicity probabilities.

The joint search found a more favorable trade-off between antimicrobial and toxicity objectives, extending the Pareto fronts toward lower predicted toxicity at comparable antimicrobial fitness (**Fig. 5b**). Across pooled seeds, it produced 642 distinct sequences on its three-objective Pareto front, compared with 227 for the APEX2-only search when evaluated on the same three objectives. The trajectories of amino-acid frequencies also differed (**Fig. 5c**). Both searches enriched lysine, but optimization with APEX2 alone strongly enriched leucine and serine. Joint optimization reduced leucine, phenylalanine and serine relative to the baseline and retained higher fractions of valine, proline and arginine. These changes show how adding toxicity objectives redirects the sequence search.

Finally, our group has developed several models through different peptide mining and design campaigns. These include ApexCPP as part of ApexDuo [22], ApexImmuno [23] and ApexFold [24], alongside the three models presented here: APEX2, ApexTox and ApexHemo. We bring these and the earlier APEX models together in the ApexSuite Python package, available under a permissive MIT license (**Fig. 5d**). ApexSuite provides a common interface for predicting antimicrobial activity, cell penetration, immunomodulation, cytotoxicity, hemolysis and secondary structure, and supports both biological sequence mining and model-guided design. This allows users to combine the relevant models within the same discovery workflow.

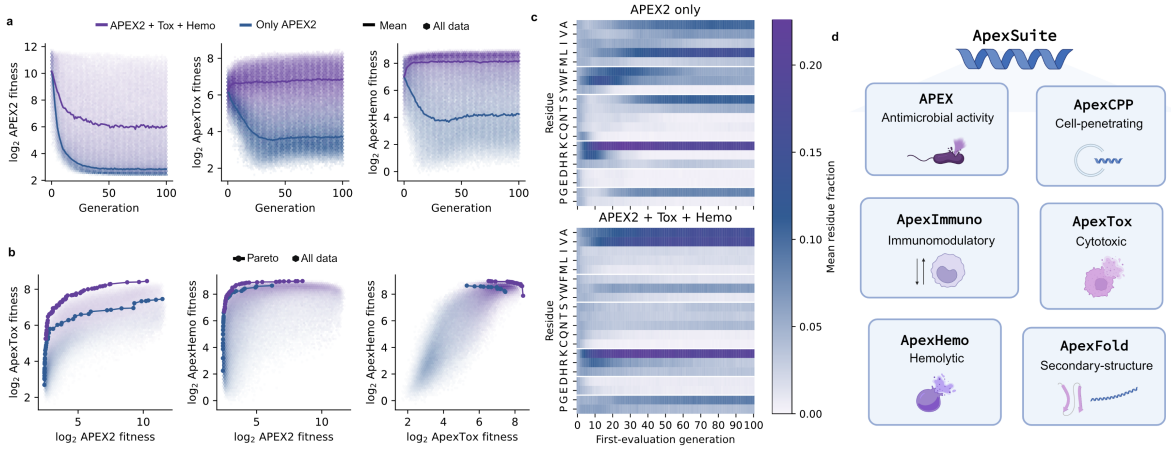


Figure 5: Joint peptide design and ApexSuite. a, Fitness for antimicrobial activity, cytotoxicity and hemolysis during APEX2-only and joint optimization. Solid lines show mean fitness and background shading shows candidate density. Lower APEX2 fitness and higher ApexTox/ApexHemo fitness are favored; toxicity scores for APEX2-only candidates were obtained after optimization. **b,** Pairwise Pareto-front comparisons. Lines connect nondominated candidates after pooling seeds and removing sequence duplicates; each front is calculated using the two displayed objectives. **c,** Residue-frequency trajectories of newly evaluated peptides. Colors show the mean fraction of each residue among peptides first evaluated in that generation. **d,** Prediction tasks combined in ApexSuite. Searches use five paired seeds, populations of 256 peptides, a peptide length of 20 residues and 100 offspring generations. Trajectory means give equal weight to each seed. Fitnesses are composite objectives expressed in $\log_2$ units.

3 Discussion

Antimicrobial peptides offer a promising avenue for the next generation of antibiotics, and both discovering peptides from natural sources and designing new unseen-to-nature sequences are ongoing efforts in this endeavor. Accurate predictive models are central to these efforts. Here we present APEX2, ApexTox and ApexHemo, which push the frontier of predictive models by combining high-quality data, formulations that reflect how activity and toxicity are measured experimentally, and modern deep learning architectures. Compared with the evaluated AMP models, APEX2 improves activity discrimination and quantitative potency prediction, offering a useful balance between identifying active peptides and limiting false positives. This balance, together with its separate activity and potency outputs, makes it truly useful during both mining and design. Similarly, ApexTox and ApexHemo improve discrimination relative to public baselines, pushing the frontier of peptide safety models.

A central design choice in these models is the combination of broad public activity labels with detailed, high-quality quantitative data from in-house experiments and carefully curated public assays. The broad labels provide supervision for learning shared sequence representations, while strain-specific antimicrobial assays and concentration-dependent toxicity measurements in the relevant sequence domain supervise the main outputs. This combination captures both broad and highly specific patterns. However, the available measurements still cover only a small fraction of peptide sequence space, and prospective evaluation across more distant sequences will be important for establishing how well these predictions generalize beyond the training distribution.

The APEX2 ablation experiments further show that the modelling formulation contributes to improved performance alongside the architectural updates. These results are intriguing and highlight the value of aligning training objectives with the information provided by the experiments. We hypothesize that this closer alignment between training targets and experimental evidence allows the models to learn a more natural representation of the data and could therefore also improve generalization to unseen peptides, a possibility that could be tested through prospective comparisons.

APEX2, ApexTox and ApexHemo, together with ApexSuite, are released under a permissive license and provide a resource that we hope will be useful to the community for continued efforts in peptide mining and design.

4 Methods

4.1 Neural network architectures

We developed APEX2, ApexTox and ApexHemo around a common sequence-encoder framework to learn how peptide sequences relate to antimicrobial activity, cytotoxicity and hemolysis. Peptides of up to 50 residues were framed by start and end tokens and padded to 52 positions. Each residue received 566 fixed AAindex descriptors [25], optionally concatenated with a trainable amino-acid embedding of 16, 32 or 64 dimensions. Alongside these residue-level inputs, models could receive a frozen 480-dimensional ESM2-35M embedding [26] and 1,132 AAindex features summarizing the whole sequence. The ESM2 embedding summarized final-layer residue representations by averaging across residues, excluding special and padding tokens; the AAindex summary concatenated the mean and standard deviation of each descriptor across residues.

We explored four encoder families: bidirectional gated recurrent units (GRUs) [27], bidirectional long short-term memory networks (LSTMs) [28], convolutional neural networks (CNNs) and

Transformer encoders [29]. Within the selected ensembles, recurrent members had one to three layers, while CNNs applied parallel convolutions with kernel widths 3, 5 and 7, followed by ReLU activation, concatenation and linear projection. Selected Transformers had two or four layers, four attention heads and GELU feed-forward blocks, without an additional positional embedding. To obtain a single peptide representation, encoder outputs passed through layer normalization and two stages of learned attention pooling, followed by linear projection. In the fitted models, Transformer self-attention masked padding keys, while recurrent encoding and the shared pooling stages operated on the fixed-length padded arrays.

When included, the optional ESM2 and AAindex sequence summaries passed through separate linear, layer-normalization, ReLU and dropout blocks before concatenation with the sequence representation. This combined representation fed separate classification and regression multilayer perceptrons, each with three hidden layers of widths D, $D/2$ and $D/4$, layer normalization, ReLU activation and dropout. Across the selected ensembles, the encoder hidden-width parameter was 64, 128 or 256, D ranged from 256 to 2,048, and dropout ranged from 0 to 0.3. For APEX2, the classification branch combined shared hidden layers with separate final linear layers to produce 11 strain-activity logits and one broad AMP logit, while the regression branch produced 11 ReLU-activated conditional-potency outputs in the native coordinate $6 - \log_{10}(\mathrm{MIC}/\mu\mathrm{M})$.

For toxicity prediction, we adapted the same classification and regression branches to describe concentration–response relationships, with the regression output representing $\log_{10}$ CC50 or HC50 in μM. An optional additional output parameterized the Hill slope as $h = \exp[\ln(4)\tanh(z)]$, constraining it between 1/4 and 4; otherwise, $h = 1$. The resulting curve provided predictions at the supplied concentrations. To account for cell context in ApexTox, models could additionally incorporate a learned 16-dimensional cell-identity embedding, projected and added to the first regression hidden layer. Cell identity entered only this branch, so the shared encoder and broad-classification branch remained sequence-only; missing or unseen identities received an unknown category with zero cell-specific contribution. Eight of the 11 selected ApexTox members used cell context and four learned the Hill slope, while six of the nine ApexHemo members learned the Hill slope.

4.2 Antimicrobial datasets and APEX2 training

To train strain-specific activity and potency outputs, we used in-house assays against *A. baumannii* ATCC 19606; *E. coli* ATCC 11775, AIC221 and AIC222; *K. pneumoniae* ATCC 13883; *P. aeruginosa* PAO1 and PA14; *S. aureus* ATCC 12600 and methicillin-resistant ATCC BAA-1556; and vancomycin-resistant *E. faecalis* ATCC 700802 and *E. faecium* ATCC 700221. Missing peptide–strain measurements were masked, while numerical MICs were represented by compatible dilution intervals and explicitly reported ranges retained their bounds. Public main-head data from DBAASP [11] supplemented these measurements, contributing 1,039 training peptides with 1,181 targets and a holdout of 200 peptides with 226 targets. These public data were also subject to the twofold dilution intervals used for training on the in-house data.

To separate model development from evaluation, we allocated peptide identities jointly across sources, assigning the in-house collection to 70% training, 20% validation and 10% test partitions while keeping identical sequences together. The allocation penalized cross-partition similarities above 90% full-global identity, although 11 above-threshold pairs remained; the training partition was further divided into three inner folds for cross-validation. Broader activity supervision came from records in DBAASP [11], APD6 [13], DRAMP [12], dbAMP [30], literature reports and reviewed UniProt fragments [17]. Source-specific label reconciliation and length-matched background sampling yielded a 49,188-peptide auxiliary dataset balanced between positive and

negative labels. This dataset was screened against the outer main holdouts for exact matches and greater-than-90% full-global identity.

We trained the broad and strain-specific activity heads with binary cross-entropy loss. Conditional potency was learned from the experimental MIC intervals by penalizing the distance of each prediction outside its interval $[l, u]$ as

$$\ell(\hat{x}; l, u) = \max(l - \hat{x}, 0)^2 + \max(\hat{x} - u, 0)^2, \tag{1}$$

with predictions and bounds expressed in $\log_2$ MIC units. The native transformed activity coordinates were rescaled to these dilution units, with only measured active assays contributing to the conditional potency loss. Cosine penalties between neighboring outputs also encouraged similar strain-specific output weights for related bacterial targets. These neighbors corresponded to the two nearest other targets for each output according to GTDB R214.1 species-representative phylogenetic distances [31], with separate coefficients controlling the activity and potency penalties. We then compared the expanded and earlier encoder families under hurdle/interval and imputed MSE training to distinguish the contributions of architecture and loss formulation.

For each architecture/loss comparison, we evaluated 300 configurations using three-fold training cross-validation. The resulting out-of-fold predictions guided a multistart add/remove/swap search that combined macro AUROC with interval concordance for hurdle models or log-MIC R^2 for MSE models. After up to 500 feasible starts, the smallest evaluated ensemble within the recorded utility tolerance was selected. The selected members were then refitted on the training data, with validation-based checkpoint selection and calibration. Across members, MIC predictions were averaged geometrically and activity probabilities arithmetically after each member's sigmoid transformation, with validation-based calibration adjusting strain probabilities and potency uncertainty.

4.3 Antimicrobial performance benchmarks

To compare activity discrimination and potency ranking across models, we evaluated their outputs against the in-house strain-specific assays. AUROC, AUPRC and interval concordance were calculated separately for each strain, with each strain contributing equally to the reported averages. For APEX2, the comparison included the strain activity, broad activity and conditional MIC heads against the same strain-specific assay labels, with the broad probability reused across strains and lower conditional MICs ranked as more active. Precision–recall curves were averaged on a common recall grid, while Prec@100 measured the active fraction among each strain's 100 highest-ranked peptides, with score ties resolved alphabetically by sequence.

For binary classification, we followed the native probability thresholds: ≥ 0.5 for the APEX2 activity heads and OmegAMP, and > 0.5 for AMPlify, amPEPpy and the HydrAMP classifiers. The corresponding cutoff for ApexPathogen was MIC ≤ 128 μM. Within each strain, observed positives provided the denominator for true-positive and false-negative rates, and observed negatives for true-negative and false-positive rates. These rates were then averaged across strains, with LR+ calculated as the ratio of the unrounded mean TPR to the unrounded mean FPR. Conditional MIC outputs contributed continuous rankings without a binary cutoff.

To assess potency ranking, we calculated concordance from definitely ordered experimental intervals, excluding overlapping intervals and giving half credit to prediction ties. For numerical error and R^2, the comparison was restricted to active assays with numerical MIC measurements, excluding explicit intervals and comparing predicted and reported MICs in $\log_2$ space. Mean-corrected MSE quantified residual error after removal of a constant prediction offset; for each model–strain pair, the mean $\log_2$ prediction error was subtracted from its predictions on the same test observations.

For comparisons with public models on the in-house assays, we constructed a shared cohort by excluding exact sequence matches to the union of available baseline reference pools and peptides whose encoded HydrAMP inputs matched reference inputs. This retained 408 validation and 195 test peptides. For models that return one score per peptide, that score served all separately labelled strain tasks: this applied to AMPlify, amPEPpy, both HydrAMP classifiers and AMP-READ. HydrAMP-MIC specifically classifies peptides by the probability of a low *E. coli* MIC. OmegAMP scores came from the matching strain classifier where available, otherwise from its species classifier and then its broad classifier. AMPGen supplied its *E. coli* or *S. aureus* output for matching species and the mean of its two log-MIC outputs for other species. Higher classifier probabilities and lower MICs or native AMP-READ scores indicated greater activity. Interval concordance in **Fig. 2b–c** and **Table 1** evaluated all models as rankers of measured active assays, including classifier scores and quantitative MIC predictions; R^2 was calculated only for quantitative MIC outputs.

The common sequence-filtered external bacterial cohort provided a test of performance beyond the in-house assays. For APEX2 and ApexPathogen, matching strain predictions took precedence; otherwise, log-MIC predictions were averaged across heads for the matching species, or across all heads if the species was not represented. Other models retained the score mappings described above. To give each species equal weight, we calculated metrics for each reported strain, averaged them within species and then averaged across species.

4.4 Cytotoxicity and hemolysis data curation

To bring together data scattered across projects, we first assembled the relevant datasets from all projects in a shared local repository. Agents orchestrated through Codex then reconciled source workbooks, plate-reader exports, plate maps, peptide lists, native plotting files and publication materials, recording any issues in a single file. Human reviewers audited these issues and recorded decisions for the agents to implement. This process continued until all identified issues had been resolved, followed by further quality control with humans and agents. These additional checks identified errors missed during the initial rounds, and curation continued until no further errors or issues could be found.

Once the records had been reconciled, we fitted equally weighted positive-dose summaries to estimate CC50 and HC50, retaining source-normalized viability and hemolysis values without universal clipping to 0–100%. Both curves had fixed plateaus at 0 and 100% and a Hill slope of one:

$$V(c) = \frac{100}{1 + c/K}, \qquad H(c) = \frac{100c}{K + c}, \tag{2}$$

where K is CC50 or HC50. A fitted crossing outside the dose range included in the fit was reported as an inequality. Following review, we excluded later reversed responses from endpoint fitting for eight hemolysis and six cytotoxicity curves, while retaining the 30 affected concentration summaries in the archive. The review also identified 31 highly atypical cytotoxicity curves for omission from default training.

Original DBAASP assays [11] extended the in-house collection with quantitative cytotoxicity and hemolysis targets. Broad classification drew on toxicity annotations from ToxinPred3 [20], DBAASP and DRAMP [12] for ApexTox, and hemolysis annotations from Hemolytik2 [32], DBAASP, DRAMP, HAPPENN [33] and HemoPI2 [21] for ApexHemo. As for the in-house data, we audited these records for outlying endpoint values and corrected some values using information from the original source studies.

4.5 Toxicity model training and evaluation

We kept evaluation peptides separate from those used for fitting by assigning all assays for a given sequence to the same partition, yielding in-house training/validation/test sets of 1,025/293/146 cytotoxicity peptides and 978/280/140 hemolysis peptides. Public quantitative assays supplemented endpoint training, while eligible binary evidence supervised broad classification. Assays that did not yield an exact 50% endpoint could still constrain its location, so they contributed finite intervals or censored limits alongside exact targets, subject to each model's source and chemistry eligibility rules.

For each ensemble member, an endpoint K and positive Hill slope h defined the concentration–response curve, with viability given by $V(c) = 100/[1 + (c/K)^h]$ and hemolysis by $H(c) = 100 - V(c)$. The slope was fixed or learned according to the selected configuration, and ApexTox additionally allowed fitted cell-line context in the main branch, with an unknown-context representation for sequence-only predictions. The endpoint loss penalized squared distance outside the log-concentration bounds. To also learn from responses across the assayed concentrations, we added a squared-error loss against responses clipped to 0–100%, normalized by the variance of the original fitting responses; validation retained the original measured responses. Concentration-response error and broad classification contributed separate losses with their weights retained for each member, and learned Hill slopes were regularized toward one with coefficient 0.1.

We selected ensembles using training out-of-fold predictions to balance endpoint accuracy and ranking, minimizing $\log_{10}(\mathrm{GMFE}) + 2(1 - C)$, where GMFE is the geometric mean fold error for exact endpoints and C is interval concordance. Selected members were then refitted with validation-based checkpoint selection. The arithmetic mean of their complete response curves defined the ensemble response, with CC50 or HC50 obtained by locating its 50% crossing through 45-step bisection.

Correlations in logarithmic concentration space quantified agreement with exact endpoints, with each peptide receiving equal total weight. Interval concordance extended the ranking comparison to bounded measurements by evaluating all models on definitely ordered censored and exact targets. Higher classifier or hybrid scores and lower predicted endpoints indicated greater toxicity. Concordance excluded pairs from the same sequence, gave half credit to prediction ties and weighted each pair by the product of the inverse assay counts of its peptides; cytotoxicity comparisons pooled the available cell contexts. For binary discrimination, experimental CC50 at or below 50 μM or HC50 at or below 128 μM defined positive labels, with ambiguous bounds and conflicting classes excluded. Cytotoxicity classes were resolved within cell context, with each peptide receiving equal total weight across contexts.

During auditing, we found that the training splits in HemoPI2 did not adhere to our standards, and that it used faulty labels in public data that were corrected. We therefore refitted HemoPI2's random-forest classifier and regressor to apply consistent HC50 curation and unit conversion and exclude training peptides overlapping or closely related to the evaluation cohorts. The official descriptor implementations and fixed model settings were retained. Public training targets were rebuilt from the source HC50 assays using the assay-curation and unit-conversion policy described above, within the intersection of our training partition and HemoPI2's original training membership. Training excluded peptides with exact sequence matches, matches above 90% full-global identity or identical first-40-residue inputs in the held-out cohorts. The classifier used 910 public training peptides with decisive binary labels at an HC50 cutoff of 100 μM.

4.6 Human-proteome scanning and known-peptide recovery

To examine model predictions in a realistic peptide-mining setting, we scanned 20,416 reviewed canonical human proteins from UniProt proteome UP000005640 [17], snapshot 2026_03. The scan generated exhaustive windows of 8–35 residues at stride one, retaining every protein occurrence and excluding windows with noncanonical residues. A reproducible sample came from 512 draws of prediction shards with replacement in proportion to occurrence counts, with 512 rows sampled per draw, using seed 42. All 262,144 occurrence draws contributed to the Spearman correlations. Composition analyses excluded repeated occurrences and exact sequence duplicates before selecting equal upper and lower 20% cohorts within each peptide length. Residue frequencies were calculated with a pseudocount of 0.5 and equal weight per length, with enrichment expressed as a $\log_2$ ratio.

We extended this analysis to host-cell toxicity by selecting 10,000 distinct peptides across peptide lengths and high, intermediate and low AMP-score groups. Antimicrobial activity was ranked by the median strain MIC divided by the median strain activity probability, while residue contrasts were matched by peptide length and AMP group. Within these comparisons, lower CC50/HC50 values and higher broad toxicity probabilities defined greater predicted toxicity.

We also assessed whether the models could recover 15 known human encrypted peptides, auditing exact training membership and containment in longer training sequences separately. Eight peptides occurred within longer APEX2 auxiliary training sequences despite having no exact training match. The MIC/probability score combined activity and potency for APEX2 recovery, with its constituent predictions retained separately. A peptide counted as recovered if its minimum score across strains was below 128; the corresponding criterion for ApexPathogen was a minimum strain MIC below 128 μM.

4.7 Multi-objective peptide design

We compared an APEX2-only search with a joint search using APEX2, ApexTox and ApexHemo to examine how toxicity predictions influence model-guided peptide design. Both searches ran with five independent seeds, paired initial populations, a population size of 256 peptides, a fixed peptide length of 20 residues and 100 offspring generations. During optimization, the per-residue mutation probability was 1/20 and the crossover probability was 0.8. Let G be the geometric mean strain MIC, $\bar{p}$ the mean strain activity probability, a broad AMP probability, and t and h broad cytotoxicity and hemolysis probabilities. Fitness was defined as

$$A = \log_2(G/\bar{p}) - 0.25\log_2(a), \qquad \text{minimize}, \tag{3}$$

$$T = \log_2(\mathrm{CC50}) + 0.25\log_2(1-t), \qquad \text{maximize}, \tag{4}$$

$$H = \log_2(\mathrm{HC50}) + 0.25\log_2(1-h), \qquad \text{maximize}. \tag{5}$$

The APEX2-only search optimized A, while the joint search retained all three as separate Pareto objectives. Probability terms in the fitness calculations were floored at 10^{-8}; inference ran with calibration disabled and an unknown cell context for ApexTox.

Each search evaluated 25,856 unique sequences per seed. Generation summaries described the newly evaluated cohorts, giving equal weight to each seed. Pooled Pareto analyses excluded exact sequence duplicates within each optimizer, with dominance recomputed using only the displayed objectives in each pairwise plot.

4.8 ApexSuite implementation

To make the predictors available through a common interface for peptide mining and design, we integrated them in ApexSuite while preserving each model's encodings, fitted features and

aggregation rules. Comparisons with the source implementations verified inference, while parameter fingerprints and settings retained in model bundles and run records supported reproduction of predictions, proteome scans and design runs.

5 Data availability

To be written.

6 Code availability

To be written.

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8 Author Contributions

E.H. designed and trained the models, ran experiments, created figures and wrote the manuscript. H.C. and W.F. contributed to model design and ideation. C.F. supervised thr project and contributed the project design and writing of the manuscript. M.D.T and J.M. contributed to project ideation.

References

[1] Lei Wang et al. "Therapeutic peptides: current applications and future directions". In: *Signal Transduction and Targeted Therapy* 7.1 (Feb. 2022). ISSN: 2059-3635. DOI: 10.1038/s41392-022-00904-4. URL: http://dx.doi.org/10.1038/s41392-022-00904-4.

[2] Brian P. Lazzaro, Michael Zasloff, and Jens Rolff. "Antimicrobial peptides: Application informed by evolution". In: *Science* 368.6490 (May 2020). ISSN: 1095-9203. DOI: 10.1126/science.aau5480. URL: http://dx.doi.org/10.1126/science.aau5480.

[3] Fangping Wan et al. "Machine learning for antimicrobial peptide identification and design". In: *Nature Reviews Bioengineering* 2.5 (Feb. 2024), pp. 392–407. ISSN: 2731-6092. DOI: 10.1038/s44222-024-00152-x. URL: http://dx.doi.org/10.1038/s44222-024-00152-x.

[4] Fangping Wan et al. "Deep-learning-enabled antibiotic discovery through molecular de-extinction". In: *Nature Biomedical Engineering* 8 (2024), pp. 854–871. DOI: 10.1038/s41551-024-01201-x.

[5] Diogo Soares et al. "OmegAMP: Targeted AMP Discovery via Biologically Informed Generation". In: *Transactions on Machine Learning Research* (2026). ISSN: 2835-8856. URL: https://openreview.net/forum?id=hAq3XLZ9ex.

[6] Marcelo D. T. Torres et al. “A generative artificial intelligence approach for peptide antibiotic optimization”. In: *Nature Machine Intelligence* 8.5 (May 2026), pp. 841–856. ISSN: 2522-5839. DOI: 10.1038/s42256-026-01237-5. URL: http://dx.doi.org/10.1038/s42256-026-01237-5.

[7] Chenkai Li et al. “AMPlify: attentive deep learning model for discovery of novel antimicrobial peptides effective against WHO priority pathogens”. In: *BMC Genomics* 23 (2022), p. 77. DOI: 10.1186/s12864-022-08310-4.

[8] Travis J. Lawrence et al. “amPEPpy 1.0: a portable and accurate antimicrobial peptide prediction tool”. In: *Bioinformatics* 37.14 (2021), pp. 2058–2060. DOI: 10.1093/bioinformatics/btaa917.

[9] Paulina Szymczak et al. “Discovering highly potent antimicrobial peptides with deep generative model HydrAMP”. In: *Nature Communications* 14 (2023), p. 1453. DOI: 10.1038/s41467-023-36994-z.

[10] Shuwen Jin et al. “AMPGen: an evolutionary information-reserved and diffusion-driven generative model for de novo design of antimicrobial peptides”. In: *Communications Biology* 8 (2025), p. 839. DOI: 10.1038/s42003-025-08282-7.

[11] Malak Pirtskhalava et al. “DBAASP V4: condition-resolved bioactivity and chemical structures for data-driven peptide and peptidomimetic discovery”. In: *Nucleic Acids Research* (Oct. 2026). ISSN: 1362-4962. DOI: 10.1093/nar/gkag1050. URL: http://dx.doi.org/10.1093/nar/gkag1050.

[12] Tianyue Ma et al. “DRAMP 4.0: an open-access data repository dedicated to the clinical translation of antimicrobial peptides”. In: *Nucleic Acids Research* 53.D1 (Jan. 2025), pp. D403–D410. ISSN: 1362-4962. DOI: 10.1093/nar/gkae1046. URL: http://dx.doi.org/10.1093/nar/gkae1046.

[13] Guangshun Wang et al. “APD6: the antimicrobial peptide database is expanded to promote research and development by deploying an unprecedented information pipeline”. In: *Nucleic Acids Research* 54.D1 (Jan. 2026), pp. D363–D374. ISSN: 1362-4962. DOI: 10.1093/nar/gkaf860. URL: http://dx.doi.org/10.1093/nar/gkaf860.

[14] Jeremy P Bradshaw. “Cationic Antimicrobial Peptides: Issues for Potential Clinical Use”. In: *BioDrugs* 17.4 (2003), pp. 233–240. ISSN: 1173-8804. DOI: 10.2165/00063030-200317040-00002. URL: http://dx.doi.org/10.2165/00063030-200317040-00002.

[15] Mark A. Hanson et al. “Prioritize validation over discovery of novel antibiotics”. In: *Proceedings of the National Academy of Sciences* 123.44 (2026), e2616439123. DOI: 10.1073/pnas.2616439123. eprint: https://www.pnas.org/doi/pdf/10.1073/pnas.2616439123. URL: https://www.pnas.org/doi/abs/10.1073/pnas.2616439123.

[16] Paulina Szymczak et al. “BATTLE-AMP: Benchmarking Antimicrobial Peptide Predictors”. In: (June 2026). DOI: 10.64898/2026.06.19.733349. URL: http://dx.doi.org/10.64898/2026.06.19.733349.

[17] The UniProt Consortium. “UniProt: the Universal Protein Knowledgebase in 2025”. In: *Nucleic Acids Research* 53.D1 (2025), pp. D609–D617. DOI: 10.1093/nar/gkae1010.

[18] Jianda Yue et al. “A Retrospective Benchmark of Quantitative Antimicrobial Peptide Activity Prediction Tools”. In: *Small Methods* (Oct. 2026). ISSN: 2366-9608. DOI: 10.1002/smtd.71137. URL: http://dx.doi.org/10.1002/smtd.71137.

[19] Beilun Wang et al. “Explainable deep learning and virtual evolution identifies antimicrobial peptides with activity against multidrug-resistant human pathogens”. In: *Nature Microbiology* 10.2 (2025), pp. 332–347. DOI: 10.1038/s41564-024-01907-3.

[20] Anand Singh Rathore et al. "ToxinPred 3.0: An improved method for predicting the toxicity of peptides". In: *Computers in Biology and Medicine* 179 (2024), p. 108926. DOI: 10.1016/j.compbiomed.2024.108926.

[21] Anand Singh Rathore et al. "Prediction of hemolytic peptides and their hemolytic concentration". In: *Communications Biology* 8 (2025), p. 176. DOI: 10.1038/s42003-025-07615-w.

[22] Angela Cesaro et al. "Design of multimodal antibiotics using deep learning". In: *bioRxiv* (2025). Preprint. DOI: 10.1101/2024.12.20.629780.

[23] Machine Biology Group. *Anti-inflammatory peptide predictor (ApexImmuno)*. Software repository. 2026. URL: https://gitlab.com/machine-biology-group-public/anti-inflammtory_peptide_predictor (visited on 11/12/2026).

[24] Marcelo D. T. Torres, Hanqun Cao, and Cesar de la Fuente-Nunez. "Peptide structural plasticity is predictable from sequence and environment". In: *bioRxiv* (2026). Preprint. DOI: 10.64898/2026.08.09.743796.

[25] Shuichi Kawashima et al. "AAindex: amino acid index database, progress report 2008". In: *Nucleic Acids Research* 36.Database issue (2008), pp. D202–D205. DOI: 10.1093/nar/gkm998.

[26] Zeming Lin et al. "Evolutionary-scale prediction of atomic-level protein structure with a language model". In: *Science* 379.6637 (2023), pp. 1123–1130. DOI: 10.1126/science.ade2574.

[27] Kyunghyun Cho et al. "Learning Phrase Representations using RNN Encoder–Decoder for Statistical Machine Translation". In: *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, 2014, pp. 1724–1734. DOI: 10.3115/v1/D14-1179.

[28] Sepp Hochreiter and Jürgen Schmidhuber. "Long Short-Term Memory". In: *Neural Computation* 9.8 (1997), pp. 1735–1780. DOI: 10.1162/neco.1997.9.8.1735.

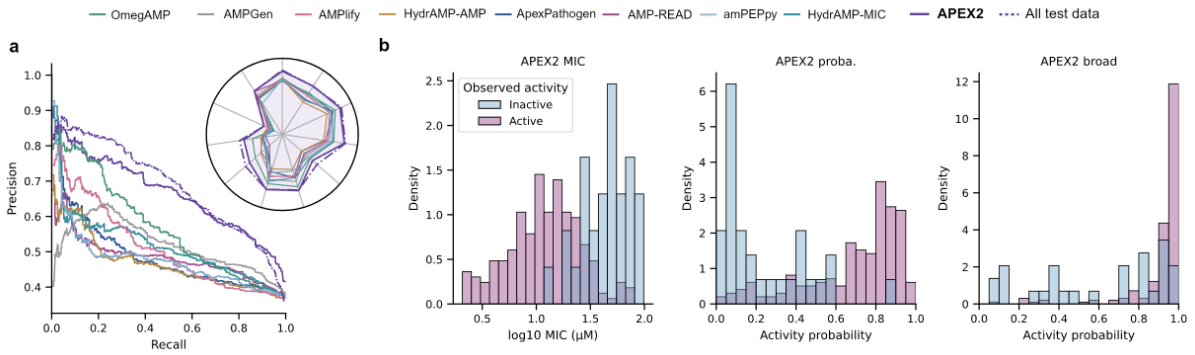
[29] Ashish Vaswani et al. "Attention Is All You Need". In: *Advances in Neural Information Processing Systems*. Vol. 30. 2017. URL: https://proceedings.neurips.cc/paper/2017/hash/3f5ee243547dee91fbd053c1c4a845aa-Abstract.html.

[30] Lantian Yao et al. "dbAMP 3.0: updated resource of antimicrobial activity and structural annotation of peptides in the post-pandemic era". In: *Nucleic Acids Research* 53.D1 (2025), pp. D364–D376. DOI: 10.1093/nar/gkae1019.

[31] Donovan H. Parks et al. "GTDB: an ongoing census of bacterial and archaeal diversity through a phylogenetically consistent, rank normalized and complete genome-based taxonomy". In: *Nucleic Acids Research* 50.D1 (2022), pp. D785–D794. DOI: 10.1093/nar/gkab776.

[32] Ayushi Singh et al. "Hemolytik 2: An Updated Database of Hemolytic Peptides and Proteins". In: *Chemical Research in Toxicology* 39.2 (2026), pp. 208–215. DOI: 10.1021/acs.chemrestox.5c00322.

[33] Patrick Brendan Timmons and Chandralal M. Hewage. "HAPPENN is a novel tool for hemolytic activity prediction for therapeutic peptides which employs neural networks". In: *Scientific Reports* 10 (2020), p. 10869. DOI: 10.1038/s41598-020-67701-3.



Extended Data Fig. 1: Precision–recall performance and public-holdout prediction distributions for APEX2. a, Precision–recall curves averaged equally across strains on a common recall grid, with per-strain average precision (AUPRC) in the radar inset. APEX2 uses its strain activity head. AMPlify, amPEPpy, both HydrAMP classifiers and AMP-READ reuse one peptide score across strains. OmegAMP uses matching strain, species or broad classifiers, in that order; AMPGen uses matching species predictions or the mean of its two log-MIC outputs. Higher classifier probabilities, lower MICs and lower native AMP-READ scores indicate greater activity. Radar axes follow the strain order in Methods, clockwise from the top, on a radial scale of 0–1. Solid curves use the shared in-house test cohort of 195 peptides; the dash-dot APEX2 curve uses all 369 in-house test peptides. AUPRC is calculated as average precision. **b,** Distributions of conditional MIC, strain-specific activity probability and broad AMP probability in the separate public holdout of 200 peptides and 226 assays, grouped by observed activity (197 active and 29 inactive assays). Histograms use all assays and are normalized separately within each activity class.

Extended Data Table 1: Peptide-level antimicrobial activity and minimum MIC prediction on the in-house test cohorts.

Model / head	AUROC	AUPRC	FPR (%)	TPR (%)	LR+	Interval C	Minimum MIC R^2
Shared test cohort: 195 peptides							
ApexPathogen	0.711	0.917	85.71	93.75	1.09	0.582	−0.642
AMPGen	0.628	0.856	—	—	—	0.675	−0.759
AMP-READ	0.650	0.879	—	—	—	0.659	—
OmegAMP	0.651	0.893	**5.71**	13.12	**2.30**	0.645	—
HydrAMP-AMP	0.523	0.855	91.43	88.75	0.97	0.606	—
HydrAMP-MIC	0.630	0.877	57.14	73.12	1.28	0.650	—
AMPlify	0.568	0.871	80.00	89.38	1.12	0.569	—
amPEPpy	0.571	0.865	80.00	90.00	1.12	0.581	—
APEX2 strain activity	**0.841**	**0.955**	40.00	91.88	**2.30**	0.748	—
APEX2 broad activity	0.727	0.907	82.86	**98.12**	1.18	0.709	—
APEX2 conditional MIC	—	—	—	—	—	**0.751**	**0.326**
Complete test cohort: 369 peptides							
APEX2 strain activity	0.882	0.938	16.24	81.35	5.01	0.754	—
APEX2 broad activity	0.850	0.908	46.15	91.27	1.98	0.720	—
APEX2 conditional MIC	—	—	—	—	—	0.769	0.400

Each peptide contributes once, with a positive label if active against at least one tested strain and a negative label if inactive against every tested strain. Classification uses the highest activity score across each peptide's assayed strains, or the general peptide score when only one output is available. AUPRC is calculated as average precision. FPR and TPR are the percentages of inactive and active peptides classified as active; LR+ is TPR/FPR calculated from unrounded rates. Binary cutoffs are probability ≥ 0.5 for APEX2 and OmegAMP, > 0.5 for AMPlify, amPEPpy and HydrAMP, and minimum predicted MIC ≤ 128 μM for ApexPathogen. No binary cutoff is established here for AMPGen or AMP-READ.

For peptides with at least one measured MIC, interval concordance (C) evaluates all models as rankers of the lowest MIC across the assayed strains. Rankings use the highest activity score across all assayed strains, equivalent to the lowest predicted MIC for quantitative outputs, without selecting outputs by observed activity. The observed minimum interval is bounded by the minima of the assay lower and upper bounds, retaining lower-edge closure. C compares definitely ordered minimum intervals and gives half credit to prediction ties. R^2 is calculated only for quantitative MIC outputs and compares predicted and reported numerical minimum MICs in $\log_2$ space, excluding peptides with explicit MIC intervals. Evaluation includes 160 peptides in the shared test cohort and 252 peptides in the complete APEX2 test cohort for both C and R^2.

The shared test cohort retains the reference-overlap exclusions and model-specific strain and species matching in **Table 1**, including the AMPGen fallback and encoded HydrAMP input exclusions, and contains 160 active and 35 inactive peptides. The complete APEX2 test cohort includes all 369 held-out test peptides, with 252 active and 117 inactive peptides. Positive-class prevalences are 0.821 and 0.683 in the shared and complete cohorts, respectively. Dashes denote metrics not evaluated for that output. Bold marks the highest value for each metric, or the lowest FPR, within the shared cohort.